



WELDING PROCEDURE SPECIFICATION (WPS)

Company Name **Renew Coil Engineering & Supply Co., Ltd.**

WPS No. **RNC-WPS-018**

Date **23-May-13**

Revision No. **0**

Supporting PQR No.(s) **RNC-PQR-018**

Date **26-Apr-13**

Type(s) **Manual**

Welding Process(es) **Gas Tungsten Arc Welding + Shield Metal Arc welding**

(Automatic, **Manual**, Machine, or Semi-Auto.)

(GTAW + SMAW)

JOINT (QW-402)

Joint Design **GROOVE & FILLET**

Backing (Yes) **SMAW** (No) **_____**

Backing Material (Type) **Weld Metal**

☐ Metal ☐ Nonfusing Metal
☐ Nonmetallic ☐ Other

Notes **_____**

See Details Attachment sheet Joint Sketch

BASE METALS (QW-403)

P-No. **15E** Group No. **1** to P-No. **15E**

Group No. **1**

Material Specification Type and Grade **A335 P91, A182 F91, A336 F91**

TO **A335 P91, A182 F91, A336 F91**

Chem. Analysis and Mech. Prop. **_____**

Thickness Range :

Base Metal : Groove **5 mm -74 mm.** Fillet **All**

Pipe Dia. Range : Groove **NPS 2 7/8"-Unlimited** Fillet **All**

Other **_____**

FILLER METALS (QW-404)

Spec. No. (SFA) **_____**

GTAW

5.28

SMAW

5.5

AWS No. (Class) **_____**

ER 90S-B9

E 9015-B9

F-No. **_____**

6

4

A-No. **_____**

5

5

Size of Filler Metals **_____**

Ø 2.4 mm

Ø 3.2 & 4.0 mm.

Deposited Weld Metals **_____**

10.0 mm.

27 mm.

Thickness Range :

Groove

20.0 mm Max.

54 mm. Max.

Fillet

ALL

ALL

Electrode-Flux (Class) **_____**

-

-

Flux Trade Name **_____**

-

-

Consumable Insert **_____**

-

-

Other **_____**

Bohler C 9MV-IGor Equivalent

Bohler FOX C 9 MVor Equivalent

POSITION (QW-405)

Position(s) of Groove **ALL**

Welding Progression **Up X Down**

Position(s) of Fillet **ALL**

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range **740[±]10°C**

Time Range **Holding Time 6 Hour**

Notes **Cooling Rate = 200°C Max.**

PREHEAT (QW-406)

Preheat Temp. Min. **200 -250°C**

Interpass Temp. Max. **300°C ±10°C**

Preheat Maintenance **Continuous at 350 °C 1 HR.**

Notes **Resistance heater**

GAS (QW-408)

Percent Composition

Shielding

Trailing

Backing

Gas(es)	(Mixture)	Flow Rate
Argon	99.99%	12-16 Ltr/Min
-	-	-
Argon	99.99%	12-16 Ltr/Min

WPS No. RNC-WPS-018 Rev. 0

ELECTRICAL CHARACTERISTICS (QW-409)

Current ☒ AC ☒ DC Polarity EN(GTAW)+EP(SMAW)
 Amps (Range) See Below Volts (Range) See Below
 Tungsten Electrode Size and Type Ø 2.4 mm , Thoriated (EWTh-2)
 (Pure Tungsten, 2% Thoriated, etc.)
 Mode of Metal Transfer for GMAW N/A
 (Spray arc, short circuiting arc, etc.)
 Electrode Wire feed speed range N/A

TECHNIQUE (QW-410)

String or Weave Bead BOTH APPLICATIONS
 Orifice or Gas Cup Size 8.0 mm - 19.0 mm
 Initial and Interpass Cleaning (Brushing, Grinding, etc.) BRUSHING OR GRINDING
 Method of Back Gouging N/A
 Oscillation N/A
 Contact Tube to Work Distance N/A
 Multiple or Single Pass (per side) MULTIPASS
 Multiple or Single Electrodes SINGLE ELECTRODE
 Travel Speed (Range) See Below
 Peening N/A
 Other N/A

Weld Layer(s)	Process	Filler Metal		Current		Volt Range	Travel Speed Range (Cm/min)	Other
		Class	Dia. (mm)	Type Polar.	Amp. Range			
1st	GTAW	ER 90S-B9	2.4	DC EN	100-130	8-12	5-10	
2nd	GTAW	ER 90S-B9	2.4	DC EN	110-140	8-12	8-12	
3 & Over	SMAW	E 9015-B9	3.2 & 4.0	DC EP/AC	90-140	27-37	5-10	

We, the undersigned, certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirement of ASME-IX, 2010 latest Revision.

Prepared by :

Mr. Kittipong M. [Signature] May 23, 2013
 Name Signed Date

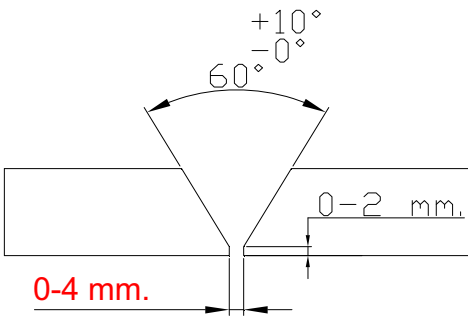
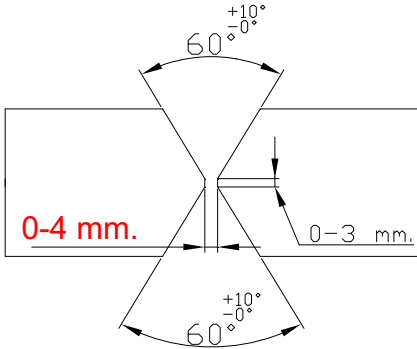
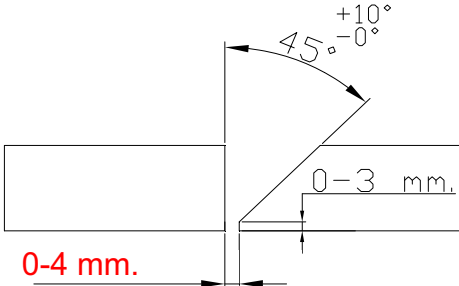
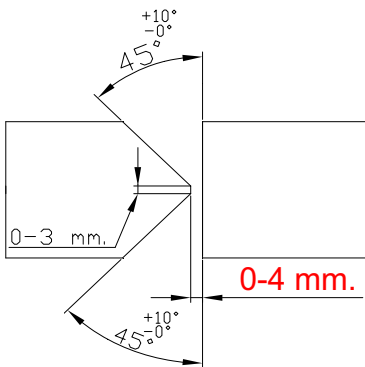
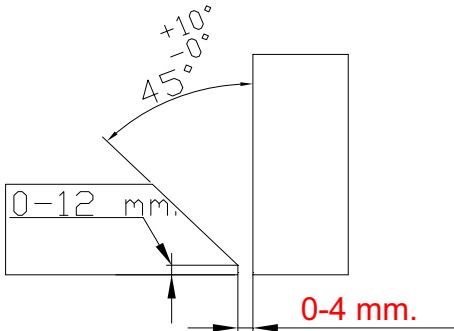
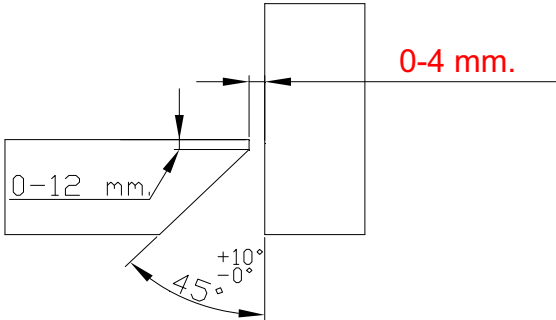
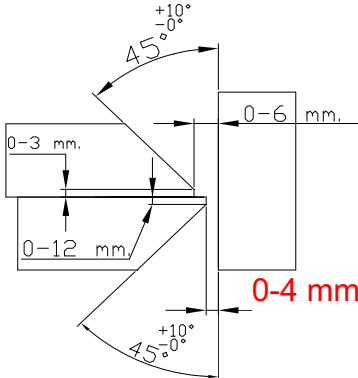
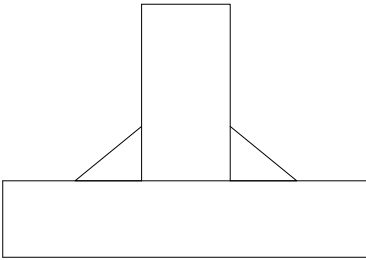
Reviewed by :

ภูชิน สีหอมไชย [Signature] 24/05/13
 Name Signed Date

Approved by :

ภูชิน สีหอมไชย [Signature] 24/05/13
 Name Signed Date

JOINT PREPARATIONS

1. Single Vee Groove 	2. Double Vee Groove 
3. Single Bevel Groove 	4. Double Bevel Groove 
5. Bevel Groove 	6. Bevel Groove 
7. Bevel Groove 	8. Fillet 



Procedure Qualification Record (PQR)

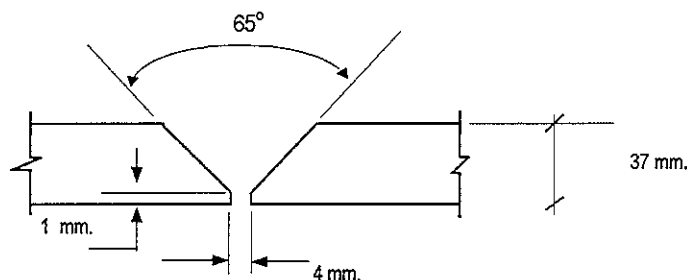
ASME-QW-200.2, Section IX, ASME Boiler Pressure Vessel Code

Record Actual Variables Used to Weld Test Coupon

RECORD OF WELDING (QW-483) PQR Number : RNC-PQR-018

Company Name :	RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.		
PQR Record Number :	RNC-PQR-018 Case 1	Date :	26 April 2013
WPS Number :	RNC-WPS-018		
Welding Process (es) :	Gas Tungsten Arc Welding (GTAW) + Shield Metal Arc Welding (SMAW)		
Type (Manual, Semi-auto, or Automatic) :	Manual		

JOINT DESIGN (QW-402)



(For combination qualifications, the deposited weld metal thickness shall be recorded for each filler metal or process used)

WELDING VARIABLES

BASE METALS (QW-403)			POSTWELD HEAT TREATMENT (QW-407)			
Material Spec.	SA 335	To	SA 335	Temperature	740° C	
Type or Grade	SA 335 P91	To	SA 335 P91	Hold Time	6 HR.	
P. No.	15E	To	15E	Others	Cooling Rate 195 °C/Hr.& Heating Rate 180° C/Hr.	
Thickness of Test Coupon		37 mm.		GAS (QW-408)		
Diameter of Test Coupon		Ø 12"				
Others						
FILLER METALS (QW-404)			Percent Composition			
				Gas (es)	Mixture	Flow Rate
			Shielding	Argon	99.95%	20 L/Min.
			Trailing	-	-	-
			Backing	Argon	99.95%	25 L/Min.
			ELECTRICAL CHARACTERISTICS (QW-409)			
SFA Specification		GTAW	SMAW	Current	DC (GTAW)	DC (SMAW)
AWS Classification		A 5.28	A 5.5	Polarity	EN	EP
Filler Metal F-No.		ER 90S-B9	E 9015-B9	Amperage	113 - 120 (GTAW)	Voltage 11 (GTAW)
Weld Metal Analysis A-No.		6	4	117 - 131 (SMAW) 23 - 24 (SMAW)		
Size of Filler Metal		5	5	Tungsten Electrode Size	Ø 2.4 mm. 2% thoriated.EWTh-2	
Others		Ø 2.4 mm.	Ø 3.2 mm.	Others	N/A	
Weld Metal Thickness		Bohler C9MV-IG	Bohler FOX C9 MV			
POSITION (QW-405)			TECHNIQUE (QW-410)			
Position of Groove		6G	Travel Speed	5.01 - 7.75 cm/min.		
Progression of Welding (Uphill or Down)		Up Hill	String or Weave Bead	Both		
Others		N/A	Oscillation	N/A		
PREHEAT (QW-406)			Multiple or single Pass		Multiple Pass	
			Multiple or Single Electrodes		Multiple Electrodes	
			Others		N/A	
Preheat Temperature		250° C	Postheat Temp.	350° C		
Interpass Temperature (max.)		300° C				
Postheat Time		1 Hr.				

Procedure Qualification Record

ASME-QW-200.2, Section IX (WPQR)

RECORD OF TEST RESULTS (QW-483) PQR Number:- RNC-PQR-018 Case-I

Tensile Test Report No.TS-13-05-050

Specimen	Width (mm)	Thickness (mm)	Area (mm ²)	Ultimate Total Load (KN)	Ultimate Strength (N/mm ²)	Type of Failure & Loc.
1264/13 TR1	19.62	30.43	597.037	446.006	747.032	WELD ZONE
1264/13 TR2	19.54	32.48	634.659	468.954	738.907	WELD ZONE

Guided Bend Tests (QW-160) Report No.BD-13-05-009

Type and Figure Number	Type of Bend	Indication	Results
1264/13 BS1	Transverse face bend	No Open Discontinuity	Accepted
1264/13 BS2	Transverse face bend	Open Discontinuity 0.74,0.5 mm.	Accepted
1264/13 BS3	Transverse root bend	No Open Discontinuity	Accepted
1264/13 BS4	Transverse root bend	No Open Discontinuity	Accepted

Toughness Tests (QW-170) Report No.IM-13-05-017

Specimen Number	Notch Location	Notch Type	Test Temp.	Impact Values	Average (Joules)	Lateral Expansion Mils
1264/13-1	Weld Metal	V-Notch	- 20 °C	7.05	8.06	
1264/13-2	Weld Metal	V-Notch	- 20 °C	10.27		
1264/13-3	Weld Metal	V-Notch	- 20 °C	6.87		
1264/13-1	HAZ	V-Notch	- 20 °C	78.31	125.23	
1264/13-2	HAZ	V-Notch	- 20 °C	159.7		
1264/13-3	HAZ	V-Notch	- 20 °C	137.69		
1264/13-1	Base Metal	V-Notch	- 20 °C	92.84	88.97	
1264/13-2	Base Metal	V-Notch	- 20 °C	71.46		
1264/13-3	Base Metal	V-Notch	- 20 °C	102.62		

Fillet Weld Test (QW-180)

Result - acceptable : Yes N/A No N/A Penetration Into Base Metal : Yes N/A No N/A
Macroetch Test Results N/A

Other Tests

Type of Test Radiographic examination results Accepted Report No.PQR-011

Hardness Testing Report No. HV-13-05-004

Macrostructure Report No. MA-13-05-002

Welders Name Mr.Rom P. / Mr.Somjai S.

Welder Number -

Tests Conducted By Mr.Keerati Sa-ngoen

Laboratory Test

LPN PLATE MILL

We certify that the statements in this record are correct and that the test coupons were prepared, welded, and tested in accordance with the requirements of section IX of the ASME Boiler and Pressure Vessel Code.

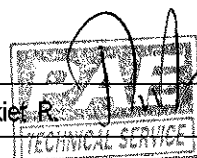
Manufacture or Contractor : _____

Inspected by : Mr.Wichit K.

Date : _____

Certified by : Mr.Chaloemkiet R.

Date : 20 May 2013



PAE Technical Service Company Limited

Public Registration No. (5) 1571/2542

69 Srinakarin Rd., suanluang, Bangkok 10250, Thailand. Tel (662) 322-022 Fax : (622) 721-2577

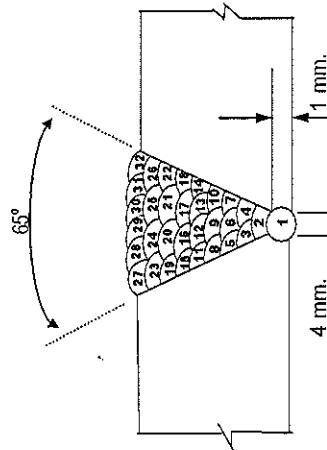
WELDER PROCEDURE QUALIFICATION RECORD

Page No. 1 of 4

Client : Renew Coils Engineering & Supply Co., Ltd.		Project Name : MINI SHUTDOWN PTTGC		Location : RNC Work Shop		Report No. PQR-006			
Case : 1		WPS No. : RNC-PQR-018		Date : 26-Apr-13					
Welder No.	Welder Name	Welding Process	Test Position	Applicable Standard	Material		Electrode Specification	Results	
					Spec	Size		Visual	RT
-	MR.ROM P.	GTAW	6G	ASME IX	A335 P 91	ø 12" Thk. 37 mm.	A5.28	ACCEPTED	ACCEPTED
-	MR.SOMJAI S.	SMAW					A5.5		

Layer (S)	Process	Filler Metal		Current		Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backing	
		Class	Dia. (mm.)	Type	Polarity				Amp	GAS	Flow Rate (l/min)	GAS
1	GTAW	ER90S-B9	2.4	DCEN		113	250	14.88	Argon	20	Argon	25
2	GTAW	ER90S-B9	2.4	DCEN		120	255	12.43	Argon	20	Argon	25
3	SMAW	E9015-B9	3.2	DCEP		117	265	33.77	-	-	-	-
4	SMAW	E9015-B9	3.2	DCEP		117	255	31.05	-	-	-	-
5	SMAW	E9015-B9	3.2	DCEP		118	251	35.86	-	-	-	-
6	SMAW	E9015-B9	3.2	DCEP		118	252	37.78	-	-	-	-
7	SMAW	E9015-B9	3.2	DCEP		118	252	34.35	-	-	-	-
8	SMAW	E9015-B9	3.2	DCEP		118	251	33.92	-	-	-	-
9	SMAW	E9015-B9	3.2	DCEP		130	254	41.05	-	-	-	-
10	SMAW	E9015-B9	3.2	DCEP		118	260	24.26	-	-	-	-

Joint Details



PAE Inspection By

Witness By

Approved By

SIGNATURE : _____
NAME : Mr. Wichit K. SERVICE
DATE : 27-Apr-2013

SIGNATURE : _____
NAME : K. N. Pong M.
DATE : 27-Apr-13

SIGNATURE : _____
NAME : ภูชิน สีทองไชย
DATE : Phuchin Seetomcha.

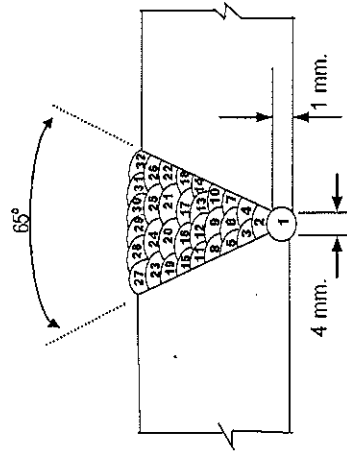
WELDER PROCEDURE QUALIFICATION RECORD

Page No. 2 of 4

Client: Renew Coils Engineering & Supply Co.,Ltd.		Project Name: MINI SHUTDOWN PTTGC		Location: RNC Work Shop		Report No. PQR-006	
Welder No.		Welder Name		Case : 1		WPS No.: RNC-PQR-018	
Welding Process		Test Position		Applicable Standard		Electrode Specification	
-		GTAW		ASME IX		A5.28	
-		SMAW				A5.5	
Results		Visual		RT		ACCEPTED	

Layer (S)	Process	Filler Metal		Current		Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backing
		Class	Dia. (mm)	Type	Polarity				Amp	GAS	
11	SMAW	E9015-B9	3.2	DCEP		130	256	31.40	-	-	-
12	SMAW	E9015-B9	3.2	DCEP		130	255	31.67	-	-	-
13	SMAW	E9015-B9	3.2	DCEP		130	253	30.14	-	-	-
14	SMAW	E9015-B9	3.2	DCEP		130	253	28.27	-	-	-
15	SMAW	E9015-B9	3.2	DCEP		131	254	29.99	-	-	-
16	SMAW	E9015-B9	3.2	DCEP		130	254	27.73	-	-	-
17	SMAW	E9015-B9	3.2	DCEP		130	252	30.19	-	-	-
18	SMAW	E9015-B9	3.2	DCEP		130	256	29.48	-	-	-
19	SMAW	E9015-B9	3.2	DCEP		130	251	31.51	-	-	-
20	SMAW	E9015-B9	3.2	DCEP		129	258	26.16	-	-	-

Joint Details



PAE Inspection By

SIGNATURE: _____
NAME: Mr. Wichit K. _____
DATE: 27-Apr-2013

Witness By

SIGNATURE: _____
NAME: K. Pong _____
DATE: 27-Apr-2013

Approved By

SIGNATURE: _____
NAME: Phuchin Seehomchai
DATE: _____

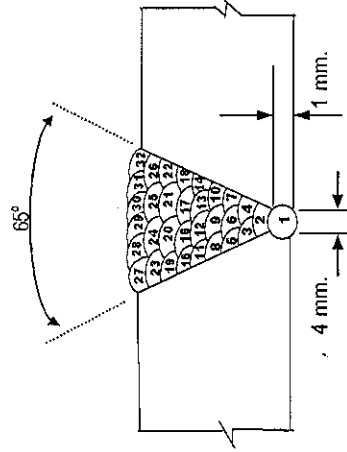
WELDER PROCEDURE QUALIFICATION RECORD

Page No. 3 of 4

Client: Renew Coils Engineering & Supply Co.,Ltd.		Project Name: MINI SHUTDOWN PTTGC		Location: RNC Work Shop		Report No. PQR-006			
Welder Name:		Case: 1		WPS No.: RNC-PQR-018		Date: 26-Apr-13			
Welder No.	Welding Process	Test Position	Applicable Standard	Material		Electrode		Results	
				Spec	Size	Specification	Visual	RT	
-	GTAW	6G	ASME IX	A335 P 91	ø 12" Thk. 37 mm.	A5.28		ACCEPTED	ACCEPTED
-	SMAW					A5.5			

Layer (S)	Process	Filler Metal		Current		Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backing	
		Class	Dia. (mm.)	Type	Polarity				Amp	GAS	Flow Rate (l/min)	GAS
21	SMAW	E9015-B9	3.2	DCEP		131	270	29.70	-	-	-	-
22	SMAW	E9015-B9	3.2	DCEP		130	261	27.65	-	-	-	-
23	SMAW	E9015-B9	3.2	DCEP		131	272	29.75	-	-	-	-
24	SMAW	E9015-B9	3.2	DCEP		130	265	25.64	-	-	-	-
25	SMAW	E9015-B9	3.2	DCEP		130	264	24.15	-	-	-	-
26	SMAW	E9015-B9	3.2	DCEP		130	250	24.28	-	-	-	-
27	SMAW	E9015-B9	3.2	DCEP		130	254	24.37	-	-	-	-
28	SMAW	E9015-B9	3.2	DCEP		130	260	25.71	-	-	-	-
29	SMAW	E9015-B9	3.2	DCEP		131	264	27.74	-	-	-	-
30	SMAW	E9015-B9	3.2	DCEP		130	249	29.62	-	-	-	-

Joint Details



PAE Inspection By

SIGNATURE: _____
NAME: Mr. Wichit K.
DATE: 27-Apr-2013

Witness By

SIGNATURE: _____
NAME: P. H. H. H.
DATE: 27-Apr-13

Approved By

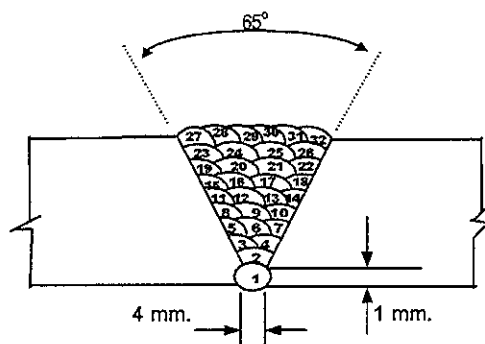
SIGNATURE: _____
NAME: Phuchin Seetomcha
DATE: _____

Page No. 4 of 4

PAE Inspection By		Witness By		Approved By	
SIGNATURE :		SIGNATURE :		SIGNATURE :	
NAME :	Mr. Wichit K. Srinire	NAME :	A. H. J. Pong	NAME :	PHUCHIN SAEHOMCHAI
DATE :	27-Apr-2013	DATE :	22/4/13	DATE :	

PAE TECHNICAL SERVICE	WELDING PROCEDURE	PQR No. : RNC-PQR-018 Case 1
		WPS No. : RNC-WPS-018
	QUALIFICATION RECORDS	THIS PQR RECORDS ACTUAL TEST CONDITION AND RESULT

Welding Sequences for RNC-PQR-018 (Butt Joint) Case - I



Pipe Dia.12" T.3.7 mm.

Weld Bead No.	Process	Class Filter Metal	Wire Size (mm)	Arc Current (Amps)	Arc Voltage (Volts)	Pot. AC/DC+	String or	Wire Feed	Rate of Travel (cm/min)	Heat Input (KJ/cm)	Gas Shielding Flow Rate (l/min)	Gas Backing Flow Rate (l/min)
							Weave Bead	Speed				
1	GTAW	ER90S-B9	2.4	113	11	DCEN	-	-	5.01	14.88	20	25
2	GTAW	ER90S-B9	2.4	120	11	DCEN	-	-	6.37	12.43	20	25
3	SMAW	E9015-B9	3.2	117	23	DCEP	-	-	4.78	33.77	-	-
4	SMAW	E9015-B9	3.2	117	23	DCEP	-	-	5.20	31.05	-	-
5	SMAW	E9015-B9	3.2	118	23	DCEP	-	-	4.54	35.86	-	-
6	SMAW	E9015-B9	3.2	118	23	DCEP	-	-	4.31	37.78	-	-
7	SMAW	E9015-B9	3.2	118	23	DCEP	-	-	4.74	34.35	-	-
8	SMAW	E9015-B9	3.2	118	23	DCEP	-	-	4.80	33.92	-	-
9	SMAW	E9015-B9	3.2	130	25	DCEP	-	-	4.75	41.05	-	-
10	SMAW	E9015-B9	3.2	118	23	DCEP	-	-	6.71	24.26	-	-
11	SMAW	E9015-B9	3.2	130	24	DCEP	-	-	5.96	31.40	-	-
12	SMAW	E9015-B9	3.2	130	24	DCEP	-	-	5.91	31.67	-	-
13	SMAW	E9015-B9	3.2	130	24	DCEP	-	-	6.21	30.14	-	-
14	SMAW	E9015-B9	3.2	130	24	DCEP	-	-	6.62	28.27	-	-
15	SMAW	E9015-B9	3.2	131	24	DCEP	-	-	6.29	29.99	-	-
16	SMAW	E9015-B9	3.2	130	24	DCEP	-	-	6.75	27.73	-	-
17	SMAW	E9015-B9	3.2	130	24	DCEP	-	-	6.20	30.19	-	-
18	SMAW	E9015-B9	3.2	130	24	DCEP	-	-	6.35	29.48	-	-
19	SMAW	E9015-B9	3.2	130	24	DCEP	-	-	5.94	31.51	-	-
20	SMAW	E9015-B9	3.2	129	24	DCEP	-	-	7.10	26.16	-	-
21	SMAW	E9015-B9	3.2	131	24	DCEP	-	-	6.35	29.70	-	-
22	SMAW	E9015-B9	3.2	130	24	DCEP	-	-	6.77	27.65	-	-



LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnmp.co.th



452

Tension Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. TS-13-05-050

Revision No. 0

Test Product	PQR No. RNC-PQR-018 (Case I)	Material Characterization	Pipe
Project Name	N/A	Test Method	ASTM A370-10 (Prepared by ASME IX : 2010)
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010 edition
Material Specification	A335 P91	Type of Test	Reduced Section Tension
Dimension	Ø 12" (37 mm Thk.)	Equipment's Capacity	200 TONS
Date Received	6 May 2013	Equipment / Serial No.	SHIMADZU UH200A / 32098926
Date Tested	8 May 2013	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Specimen Dimension					Yield		Ultimate		Elongation (%)	Remark
		Diameter (mm)	Thickness (mm)	Width (mm)	Area (mm ²)	Gauge Length (mm)	Load (kN)	Strength (N/mm ²)	Load (kN)	Strength (N/mm ²)		
1	1264/13 TR1	N/A	30.43	19.62	597.037	N/A	N/A	N/A	446.006	747.032	N/A	WELD ZONE
2	1264/13 TR2	N/A	32.48	19.54	634.659	N/A	N/A	N/A	468.954	738.907	N/A	WELD ZONE

The unit of N/mm² is equivalent to MPa

Additional details : WPS No. RNC-WPS-021

Welding Process : GTAW+SMAW

Filler/Electrode Spec't : ER90S-B9+E9015-B9

Reported by :

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 20 / 05 / 13

Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 20 / 05 / 13

Notes 1. The above results are valid exclusively for tested samples as mentioned in this report.

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4. This report is tested for LPN Metallurgical Research Center (Thailand) Co.,Ltd.

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Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnmp.co.th

TESTING
No. 0154

455

Bend Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. BD-13-05-009

Revision No. 0

Test Product	PQR No. RNC-PQR-018 (Case I)	Material Characterization	Pipe
Project Name	N/A	Test Method	ASME Sec. IX : 2010 edition
Project No.	N/A	Standart Of Test	ASME Sec. IX : 2010 edition
Material Specification	A335 P91	Type of Jig	Guided-Bend Roller Jig
Dimension	Ø 12" (37 mm Thk.)	Diameter of Meudrel	40 mm
Date Received	6 May 2013	Bend Angle	180 Degree.
Date Tested	8 May 2013	Equipment / Serial No.	SHIMADZU UH200A/32098926

Item No.	Test No.	Type of Bend	Size of Specimen (ThickxWidthxLength)	Indication	Location of Indication	Remark
1	1264/13 BS1	Transverse side bend	10.14x29.57x162.60 mm	No Open Discontinuity	N/A	-
2	1264/13 BS2	Transverse side bend	9.98x29.74x160.08 mm	Open Discontinuity	WELD ZONE	0.74, 0.5 mm
3	1264/13 BS3	Transverse side bend	10.00x29.68x160.83 mm	No Open Discontinuity	N/A	-
4	1264/13 BS4	Transverse side bend	10.27x29.71x162.32 mm	No Open Discontinuity	N/A	-

Additional details : WPS No. RNC-WPS-021

Welding Process : GTAW+SMAW

Filler/Electrode Spec't : ER90S-B9+E9015-B9

Reported by :

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 20 / 05 / 13

Approved by :

(Mr.Pavaref Preedawiphat)

Deputy Managing Director

Date 20 / 05 / 13

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Testing Laboratory LPN Plate Mill Public Company Limited

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Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnmp.co.th



0458

Impact Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. IM-13-05-017

Revision No. 0

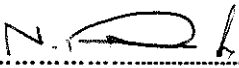
Test Product	PQR No. RNC-PQR-018 (Case I)	Material Characterization	Pipe
Project Name	N/A	Test Method	ASTM A370-10 and ASTM E23M-07a ^{el}
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010 edition
Material Specification	A335 P91	Type of Test	Charpy V-Notch Impact
Dimension	Ø 12" (37 mm Thk.)	Equipment / Serial No.	Tinius Olsen / 221811
Date Received	6 May 2013	Equipment 's Capacity	542 Joules
Date Tested	8 May 2013	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Test Temperature (°C)	Specimen Dimension			Results		Location of Test	Remark
			Thickness (mm)	Width (mm)	Length (mm)	Energy Absorbed (Joules)	Average (Joules)		
1	1264/13-1	-20°C	10.009	10.011	54.88	78.31	125.23	HAZ	-
2	1264/13-2	-20°C	10.007	10.014	54.86	159.70			
3	1264/13-3	-20°C	10.012	10.008	54.81	137.69			
4	1264/13-1	-20°C	10.005	10.008	54.73	92.84	88.97	Base Metal	-
5	1264/13-2	-20°C	10.002	10.010	54.71	71.46			
6	1264/13-3	-20°C	10.009	10.011	54.75	102.62			
7	1264/13-1	-20°C	10.011	10.011	54.90	7.05	8.06	Weld Metal	-
8	1264/13-2	-20°C	10.012	10.009	54.80	10.27			
9	1264/13-3	-20°C	10.010	10.012	54.80	6.87			

Additional details : WPS No. RNC-WPS-021

Welding Process : GTAW+SMAW

Filler/Electrode Spec't : ER90S-B9+E9015-B9

Reported by : 
 (Mr.Keerati Sa-ngoeng)
 (Acting) General Manager - Quality Assurance
 Date 20 / 5 / 13

Approved by : 
 (Mr.Pavaret Preedawiphat)
 Deputy Managing Director
 Date 20 / 5 / 13

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LPN PLATE MILL



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Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th

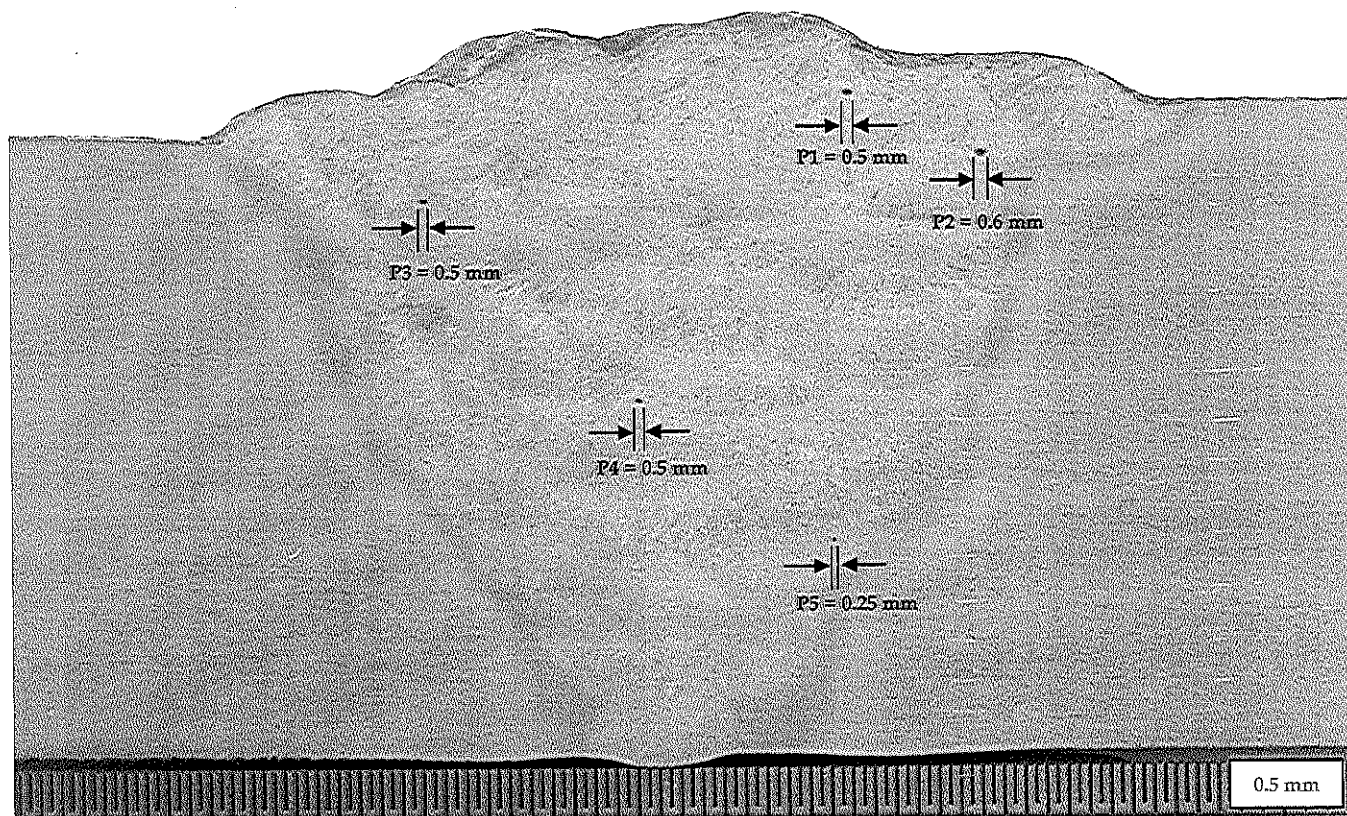
Macrostructure Analysis Report**Customer : RENEWCOILS ENGINEER & SUPPLY CO.,LTD.****Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.**

Page 1 of 1

Report No. MA-13-05-002

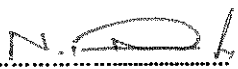
Revision No.0

Test Product	PQR No.RNC-PQR-018(case I)	Material Characterization	Pipe to Pipe
Test No.	1264/13 MA	Equipment used	Microscope upto 40X (times)
Test Location	-	Serial No.	LEICA 0650854912
Material Specification	A335P91	Method of Preparation	ASME Section IX : 2010 edition
Dimension	Ø12" (37 mm Thk.)	Test Method	ASME Section IX : 2010 edition
Welding Process/Position	GTAW+SMAW	Standard of Test	ASME Section IX : 2010 edition
Type of Joint	Butt Joint	Etchant Reagent	25 ml HNO ₃ + 75 ml H ₂ O
Welder Name / No.	N/A	Time of Etching	5 minutes
Date Received	6 May 2013	Date Tested	7 May 2013



Macro-etch Examination : The macrostructure shows porosities (P1, P2, P3,P4,P5).

Additional details :-

Reported by : 
 (Mr.Keerati Sa-ngoen)
 (Acting) General Manager - Quality Assurance
 Date : 22/05/13

Approved by : 
 (Mr.Pavaret Freedawiphat)
 Deputy Managing Director
 Date : 22/05/13

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LPN PLATE MILL



Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

13330

LPN PLATE MILL

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnrm.co.th

Hardness Testing Report

Page 1 of 1

Customer : RENEWCOILS ENGINEER & SUPPLY CO.,LTD.

Report No. HV-13-05-004

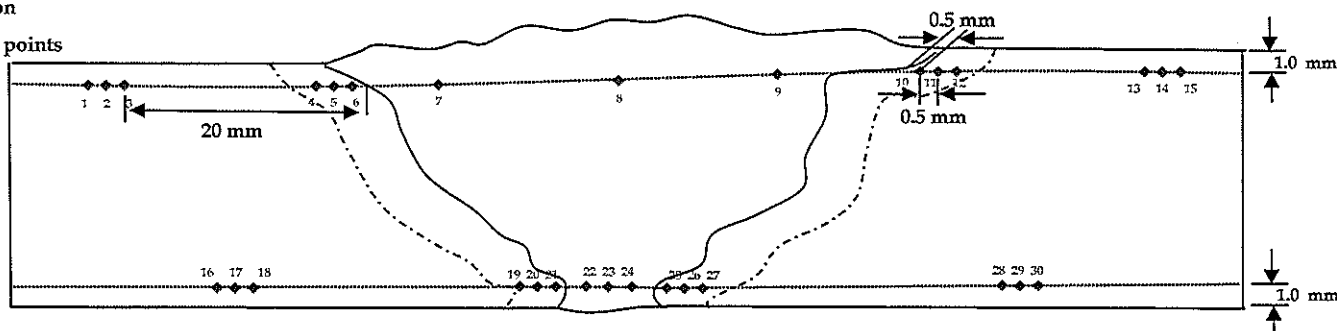
Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Revision No. 0

Test Product	PQR No.RNC-PQR-018(case I)	Material Characterization	Pipe to Pipe
Test No.	1264/13 HN	Test Method	ASTM E92-82 (Reapproved 2003) ^{E2}
Project Name	N/A	Standard of	ASME section IX: 2010 edition
Project No.	N/A	Type of test	Vicker Hardness Test
Material Specification	A335P91	Equipment	Wilson Wolpert , 432 SVD (V32D219)
Dimension	Ø12" (37 mm Thk.)	Indenter Size	Pyramid diamond, angle of 136°
Welding Process/Position	GTAW+SMAW	Test Load	HV 10 Kgf.
Type of Joint	Butt Joint	Total Force Dwell Time	15 s
Date Received	6 May 2013	Cal. Block	No. A60023 No. A62072
Date Tested	8 May 2013	Ambient Temperature	27.1 °C

Test Location

Total = 30 points



Point No.	Location	Hardness value (Scale HV)	Point No.	Location	Hardness value (Scale HV)	Point No.	Location	Hardness value (Scale HV)
1	BASE	227.8	16	BASE	233.4	*****		
2	BASE	229.0	17	BASE	232.4			
3	BASE	222.0	18	BASE	231.7			
4	HAZ	214.8	19	HAZ	238.4			
5	HAZ	227.3	20	HAZ	239.8			
6	HAZ	238.0	21	HAZ	241.3			
7	WELD	233.9	22	WELD	251.0			
8	WELD	231.2	23	WELD	252.4			
9	WELD	231.8	24	WELD	252.6			
10	HAZ	236.2	25	HAZ	257.9			
11	HAZ	220.3	26	HAZ	257.1			
12	HAZ	213.8	27	HAZ	256.9			
13	BASE	224.0	28	BASE	226.2			
14	BASE	223.4	29	BASE	227.7			
15	BASE	222.3	30	BASE	228.9			

Additional details : -

Reported by :
(Mr.Keerati Sa-ngoen)
(Acting) General Manager - Quality Assurance
Date : 20/05/13

Approved by :
(Mr.Pavaret Preeda-wiphat)
Deputy Managing Director
Date : 20/05/13

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LPN PLATE MILL



GLOBAL HEAT TREATMENT (THAILAND) CO.,LTD.

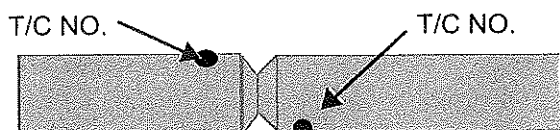
PREHEAT REPORT

CLIENT:	RENEW COILS ENGINEERING & SUPPLY CO.,LTD.		
PROJECT:	MINI SHUT DOWN PTTGC 2013		
CONTRACTOR:	GLOBAL HEAT TREATMENT (THAILAND) CO.,LTD.		
PREHEAT METHOD:	☐ GAS	☒ ELECTRIC	COMPLETION DATE : 26-04-2013
LOCATION:	☒ SHOP	☐ FIELD	REPORT NO. PH-001
RECORDER NO:	EH- 07ZA194		PAGE NO. 1
CHART SPEED:	☒ 25 MM/Hr	☐ 50 MM/Hr	☐ 100 MM/Hr

RESULT RECORDER CHART TO BE ATTACHED

DESCRIPTION	REQUEST	ACTUAL
1) PREHEAT TEMP (C° MIN)	250	250
2) INTERPASS TEMP (C° MAX)	300	300
3) POSTHEAT TEMP (C°)	350	350
4) POSTHEAT TIME (Hr)	1 HR.	1 HR.

MATERIAL : **A335 P91**

[illegible]

TOTAL	1	JOINTS.
TOTAL	12	DB.

CHECKED BY GLOBAL HEAT TREATMENT (THAILAND)	APPROVED BY	APPROVED BY
SIGN: 	SIGN: 	SIGN:
NAME: KOMKHUN D.	NAME: Mr. Chatchaman S.	NAME:
DATE: 26-04-2013	DATE: 29/04/13	DATE:

		GLOBAL HEAT TREATMENT[THAILAND] CO.,LTD. HEAT TREATMENT REPORT	
Client :	RENEW COILS ENGINEERING & SUPPLY CO.,LTD.		Page : 1
Contractor :	GLOBAL HEAT TREATMENT[THAILAND] CO.,LTD		Report No. : PW-002
Project :	MINI SHUT DOWN PTTGC 2013		Project No. :
Description :	☒ -PWHT ☐ -Normalize ☐ -Annealing ☐ -Stabilize ☐ -Solution ☐ -Dry-out		
Location :	☒ -Shop ☐ -Field		Recorder No.: EH-07ZA194
Method by:	☒ -Electric ☐ -Gas Burner ☐ -Oil burner		
Method Type:	☒ -Local ☐ -Furnace ☐ -Internal Firing		Unit:
Chart Speed :	☒ -25 mm/Hr. ☐ -50 mm/Hr. ☐ -100 mm/Hr.		
Completion Date:	28-04-2013		Procedure No.

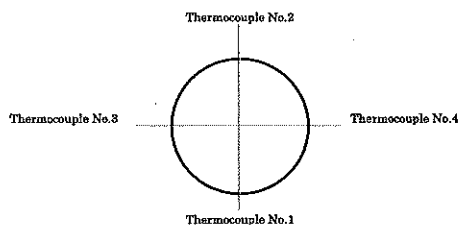
HEAT TREATMENT CONDITION/RESULTS

Description	Loading Temp[1] (°C)	Heating Rate[2] (°C/Hr.)	Holding Temp[3] (Min-Max.)	Holding Time[4] (Hr.)	Cooling Rate[5] (°C/Hr.)	Unloading Temp[6] (°C)
Request	100	200	740±15	6 HRS.	200	100
Actual	100	180	740	6 HRS.	195	100
Request						
Actual						
Request						
Actual						

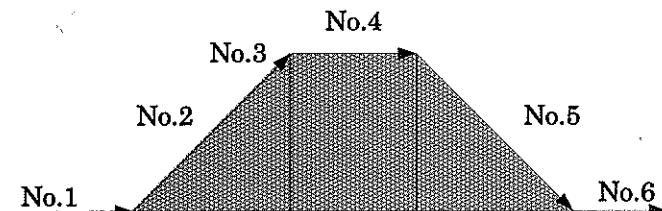
DESCRIPTION

Line No./Dwg.No./Class No./Sht.no./Spool No.	Weld No.	Size(")	THK.(MM)	Weld Type	T/C No.	Remark
RNC-WPS-018 (CASE 1)		12"	37	BW	1,2,3	
	Total:	1		Joint.	Total:	12 DB.
	Total:			EA.	Total:	Kg.

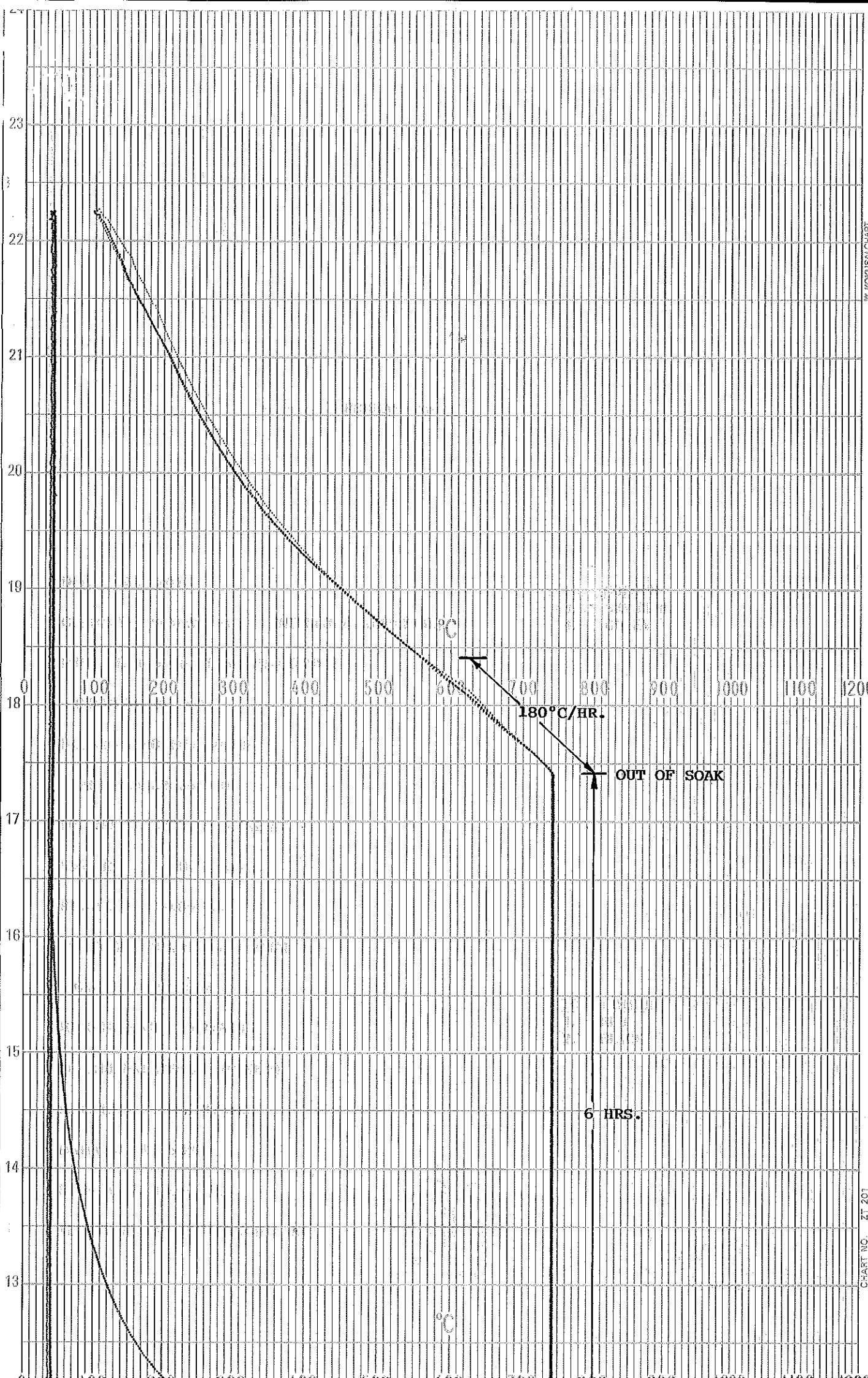
THERMOCOUPLE LOCATION

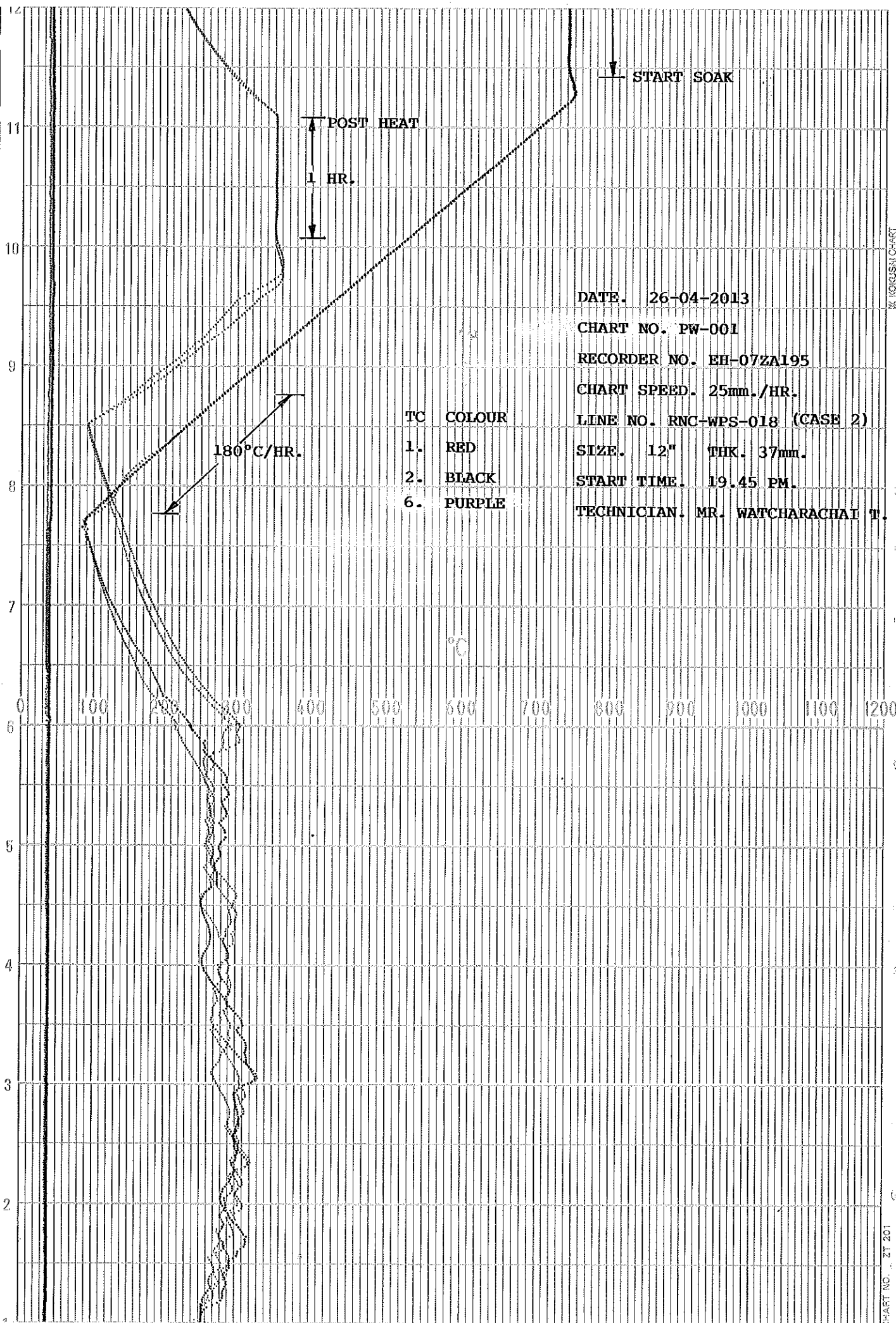


HEAT TREATMENT PROFILE



Global Heat Treatment[Thailand] Co., Ltd.					
Signature:	Checked By	Approve By	Approved by	Approved by	
Name:	KOMKHUN D.				
Date:	28-04-2013				





DATE. 26-04-2013

CHART NO. PW-001

RECORDER NO. EH-07ZA195

CHART SPEED. 25mm./HR.

LINE NO. RNC-WPS-018 (CASE 2)

SIZE. 12" THK. 37mm.

START TIME. 19.45 PM.

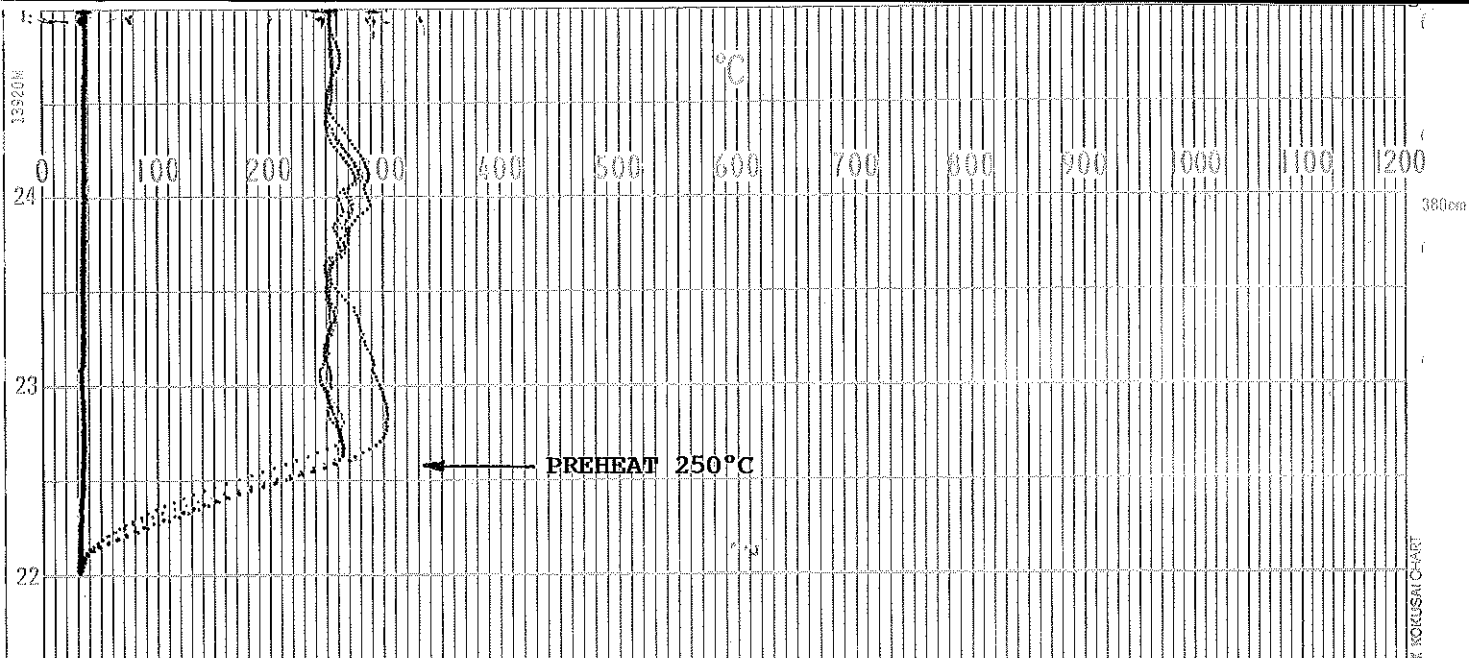
TECHNICIAN. MR. WATCHARACHAI T.

TC COLOUR

1. RED

2. BLACK

6. PURPLE



DATE. 26-04-2013

CUSTOMER. RENEW COILS ENGINEERING & SUPPLY CO.,LTD.

PROJECT. MINI SHUT DOWN PTTGC 2013

- CHART NO. PH - 001

RECORDER NO. EH 07ZA194

CHART SPEED. 25mm./HR.

LINE NO. RNC-WPS-018 (CASE 1)

SIZE. 12" THK. 37 mm.

START TIME. 10.00 AM.

TECHNICIAN. MR. WATCHARACHAI T.

- CHART NO. PH - 002

RECORDER NO. EH 07ZA195

LINE NO. RNC-WPS-018 (CASE 2)

SIZE. 12" THK. 37 mm.

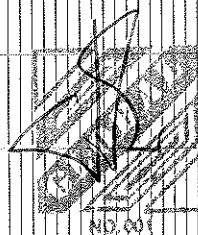
MATERIAL. A335 P91

START TIME. 10.00 AM.

TECHNICIAN. MR. CHALERMCHAI NG.

TC	COLOUR
3.	SKY BLUE
4.	GREEN

TC	COLOUR
1.	RED
2.	BLACK



Renew

PAE TECHNICAL SERVICE CO.,LTD.

NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

TEL : 02) 322-0222 , FAX : 02) 721-2577 , RAYONG Tel : 038)682208-9, Fax : 038)682206, SRIRACHA : 038)330220, 330229

Page : 1 Of 1[illegible]